Solutions to linear systems

1 Describe how the graphical solution of a system of two linear equations is found.

2 Describe how to check whether a given ordered pair is a solution of a system of equations.

3 Determine whether $(5, 2)$ is a solution of the following system of equations:

$$\begin{aligned} x + y &= 7 \\ x - y &= 3 \end{aligned}$$

4 Determine whether $(4, 17)$ is a solution of the following system of equations:

$$\begin{aligned} y &= 6x - 7 \\ 4x + 3y &= 67 \end{aligned}$$

5 Determine whether $(4, -3)$ is a solution of the following system of equations:

$$\begin{aligned} 2x + 7y &= -13 \\ 3x + 8y &= -4 \end{aligned}$$

6 Determine whether $(2, -2)$ is a solution of the following system of equations:

$$\begin{aligned} 2x + 3y &= -2 \\ 4x + 3y &= 5 \end{aligned}$$

7 Determine whether each ordered pair is a solution of the system of linear equations:

$$\begin{aligned} 4x &= 19 - y \\ x - 3y &= -5 \end{aligned}$$

a $(5, -1)$ **b** $(4, 3)$

8 Consider the system of equations:

$$\begin{aligned} y &= x - 8 \\ y &= -2x + 1 \end{aligned}$$

Determine if the following points satisfy the system:

a $(3, -5)$ **b** $(6, -2)$ **c** $(4, -4)$

9 Determine whether each ordered pair is a solution of the system of linear equations:

$$\begin{aligned} 4y &= 5x - 13 \\ 5x - y &= 22 \end{aligned}$$

a $(5, 3)$ **b** $(7, 13)$

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10 Determine whether each ordered pair is a solution of the system of linear equations:

$$y = 5x - 4$$
$$y = -x + 20$$

a $(4, 16)$ **b** $(2, 18)$ **c** $(3, -11)$

11 Determine whether each ordered pair is a solution of the system of linear equations:

$$x + 4y = 10$$
$$x - y = -5$$

a $(3, 2)$ **b** $(2, 2)$ **c** $(-2, 3)$

Table of values method

12 Consider the equations $y = 2x$ and $y = 28 - 2x$ which have the following tables of values:

$y = 2x$

x	3	4	5	6	7
y	6	8	10	12	14

$y = 28 - 2x$

x	3	4	5	6	7
y	22	20	18	16	14

State the values for x and y which satisfy the equations $y = 2x$ and $y = 28 - 2x$ simultaneously.

13 Consider the equations $y = -2x$ and $y = -12 - 4x$ which have the following tables of values:

$y = -2x$

x	-8	-7	-6	-5	-4
y	16	14	12	10	8

$y = -12 - 4x$

x	-8	-7	-6	-5	-4
y	20	16	12	8	4

State the values for x and y which satisfy the equations $y = -2x$ and $y = -12 - 4x$ simultaneously.

14 Consider the equations $x + y = 3$ and $y = x - 5$ which have the following tables of values:

$x + y = 3$

x	4	5	6
y	-1	-2	-3

$y = x - 5$

x	4	5	6
y	-1	0	1

State the values for x and y which satisfy the equations $x + y = 3$ and $y = x - 5$

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15 Consider the equations $y = 3x$ and $y = -35 - 4x$ which have the following tables of values:

$y = 3x$

x	-5	-3	-1
y	-15	-9	-3

$y = -35 - 4x$

x	-5	-3	-1
y	-15	-23	-31

State the values for x and y which satisfy the equations $y = 3x$ and $y = -35 - 4x$ simultaneously.

16 Consider the equations $y = 4x - 11$ and $y = -2x + 13$ which have the following table of values:
State the values for x and y which satisfy the equations $y = 4x - 11$ and $y = -2x + 13$ simultaneously.

x	$y = 4x + 11$	$y = -2x + 13$
2	-3	9
3	1	7
4	5	5
5	9	3
6	13	1

17 Consider the equations $y = 3x$ and $y = 15 - 2x$.

a Complete the table of values for the equation $y = 3x$.

x	-1	0	1	2	3
y					

b Complete the table of values for the equation $y = 15 - 2x$.

x	-1	0	1	2	3
y					

c Hence find the values for x and y which satisfy the equations $y = 3x$ and $y = 15 - 2x$ simultaneously.

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18 Consider the equations $y = -4x$ and $y = -4 - 5x$.

a Complete the table of values for the equation $y = -4x$.

x	-7	-6	-5	-4	-3
y					

b Complete the table of values for the equation $y = -4 - 5x$.

x	-7	-6	-5	-4	-3
y					

c Hence find the values for x and y which satisfy the equations $y = -4x$ and $y = -4 - 5x$ simultaneously.

19 Use the table of values to find the pair of x and y values that satisfy both $x + y = 5$ and $y = 2x - 1$.

a Complete the following tables:

$x + y = 5$

x	1	2	3
y			

$y = 2x - 1$

x	1	2	3
y			

b What values of x and y satisfy both $x + y = 5$ and $y = 2x - 1$?

20 Consider the equations $y = 5x$ and $y = 9 - 4x$.

a Complete the table of values for the equation $y = 5x$.

x	1	3	5
y			

b Complete the table of values for the equation $y = 9 - 4x$.

x	1	3	5
y			

c Hence find the values for x and y which satisfy the equations $y = 5x$ and $y = 9 - 4x$ simultaneously.

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21 Consider the equations $y = 3x - 1$ and $y = -5x + 23$.

a Fill in the y-values for each of the x-values given in the table:

x	$y = 3x - 1$	$y = -5x + 23$
1		
2		
3		
4		
5		

b Hence find the values for x and y which satisfy both the equations $y = 3x - 1$ and $y = -5x + 23$.

22 Consider the equations $y = -2x - 8$ and $y = 5x + 34$.

a Fill in the y-values for each of the x-values given in the table:

x	$y = -2x - 8$	$y = 5x + 34$
-7		
-6		
-5		
-4		

b Hence find the values for x and y which satisfy both the equations $y = -2x - 8$ and $y = 5x + 34$.

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Applications

23 Two phone plans have a monthly flat fee including 8 GB of data plus a cost for data overage, which is the amount of data used in excess of the included data. The costs, as shown in the table below, are calculated by which data threshold you reach. When the amount of data overage is between two threshold amounts, you are charged according to the lower threshold.

Data overage	0 GB	1 GB	2 GB	3 GB	4 GB	5 GB
BZZ($)	30	41	52	63	74	85
Telyou($)	50	56	62	68	74	80

a For which of the amounts of overage does it not matter which phone plan is chosen?

b What is the cost of each plan when they are the same price?

c If your amount of overage is always less than 1 GB, which of the plans will be the cheaper option?

24 A yoga studio has two options for people to attend their classes. One option is to purchase an annual membership that gives you unlimited access to classes for $\$840$. The other option is to pay a 'drop in' fee of $\$21$ per class.

a Complete the table below:

Number of classes attended in one year	Cost for the Annual Membership ($)	Cost to Drop in for every class($)
0		
10		
20		
30		
40		
50		

b How many times would you have to drop into a class in one year to be paying the same amount as an annual subscription?

c Which of the options should you choose if you plan on attending a class once per week, for a whole year (52 weeks), Drop in or Annnual Membership?

d Which of the options should you choose if you plan on attending a class once every two weeks for a whole year, Annnual Membership or 'Drop in' to every class?

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25 Dave obtained quotes from two plumbers:

- Plumber A charges $\$92$ for a callout fee plus $\$17$ per hour.
- Plumber B charges $\$20$ for a callout fee plus $\$35$ per hour.

a Complete the table:

Hours of work	Amount charged by plumber A	Amount charged by plumber B
4		
5		
6		
7		

b How many hours would the job have to take for plumber A and plumber B to charge the same amount?

26 Frank makes ceramic bowls and sells them online. To use a kiln for making these bowls, there is a flat fee of $\$400$ per month plus $\$25$ per bowl. He then sells the bowls for $\$50$ each.

a Complete the table:

Number of bowls made in one month	Cost to make (dollars)	Revenue from sales (dollars)
0		
4		
8		
12		
16		
20		

b How many bowls does Frank need to sell to exactly cover his costs?

c What are the total costs of making the bowls when Frank breaks even?

d Would Frank make a profit or a loss in selling 20 bowls?

e If Frank wanted to break even after making only 8 bowls, how much would he have to charge his customers for each bowl?

27 Gwen has a total of $\$11500$, which she has put into two accounts. She has four times as much money in her savings account as in her checking account. Let x represent the amount in her checking account and let y represent the amount in her savings account. Write a system of two equations that describes all the information provided. You do not need to solve the system.

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